

This module contributes to mapping the incoming simulation data to hospital patient records and ensuring data integrity. Incoming records are represented as IncomingDataPoint objects (they store in this order patient Id, measurement value, measurement type, timestamp).

HospitalPatient holds a read-only copy list of a patient's records used for safe retrieval and comparison. The PatientIdentifier looks up the HospitalPatient's records by patient Id, and the RecordComparison checks if the incoming data point exists already in the patient's records to avoid storing duplicates.

Duplicate patient records means that the original patient record and the incoming record have the same patient Id, same measurement value, same measurement type and same timestamp.

The IdentityManager manages and see through the whole identification process:

- It copies hospital patients for short term use (and for data retrieval)
- Validates whether a simulator id matches an existing hospital patient
- Retrieves the Hospital Patient
- Documents mismatches through MismatchHandler
- During adding data process adds the incoming record to the correct hospital patient if the patient exists, otherwise it manages(through DataStorage) to create a new patient and add the data to him/her.

The MismatchHandler makes documentation of unmatched simulator patient ids and the current timestamp for test, debugging, safety and monitoring purposes.